WOO YOUNG CHO

BIOLOGICAL MACHINE LEARNING · SPATIAL PROTEOMICS · COMPUTATIONAL BIOLOGY

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RESEARCH PROFILE

Computational biologist with an M.Sc. in Bioinformatics and research experience across spatial proteomics, histopathology, transcriptomics and metagenomics. Adapt, train and evaluate machine-learning models, develop biological data representations, and build reproducible analysis software. Research interests include multimodal representation learning and computational methods that connect complementary measurements of cellular and tissue states.

EDUCATION

M.Sc. Bioinformatics - Technical University of Munich and Ludwig Maximilian University of Munich

Joint programme · Oct 2021-Aug 2023 · Final grade: 1.9; thesis: 1.0 (German scale, 1.0 best)

Thesis: *Metagenomic Assembled Graphs: A novel data structure combining positional and sequential information for functional analysis on large metagenomic datasets.*

Advisors: Melanie Schirmer and Shen Jin.

Selected coursework: Machine Learning (8 ECTS); Introduction to Deep Learning (6); Advanced Data Handling and Visualization Techniques (6); Protein Prediction II for Bioinformaticians (8).

B.Sc. Bioinformatics - Technical University of Munich and Ludwig Maximilian University of Munich

Joint programme · Sep 2016-Aug 2021

Thesis: *Development and Evaluation of EnzymeFinder: Database Dependent Identification Search Algorithm to Identify Substrate-specific Biocatalysts.*

Advisors: Iris Antes and Okke Melse.

RESEARCH AND PROFESSIONAL EXPERIENCE

Arontier - Researcher, AI Team

Oct 2025-present

Seoul, Republic of Korea

Research role under South Korea's Technical Research Personnel programme (mandatory alternative military service).

- Develop computational workflows connecting multiplex tissue images, cellular measurements and patient information.
- Develop image-processing, cellular-analysis and phenotype-review workflows, with attention to reproducibility and review.
- Analyse transcriptomic data to develop and refine biological and therapeutic hypotheses with collaborators.

ConnectaGen - Bioinformatics Data Engineer / Senior Researcher

Sep 2023-Sep 2025

Seoul, Republic of Korea

Research role under South Korea's Technical Research Personnel programme (mandatory alternative military service).

- Developed genomic features and a Random Forest workflow for epidemiologically defined high-risk *E. coli*, including model evaluation and feature interpretation; second author of the resulting *Journal of Microbiology* publication.
- Led development and deployment of **ProofChase**, a Dockerized Snakemake WGS workflow for pathogen identification, resistance/virulence analysis, typing and reporting, in collaboration with governmental institutions.
- Developed and maintained GenoMycAnalyzer workflows, including detection-database curation and parameter calibration; built influenza-surveillance workflows and automated reports.
- Integrated multiple servers into a SLURM-managed environment and administered Linux access, storage, backups and job queues for bioinformatics workloads.

Technical University of Munich, AG Schirmer - Student Research Assistant (HiWi)

May-Sep 2023

Freising, Germany

Related academic research: supervised project and M.Sc. thesis, 2022-2023

- Contributed to MetaGEAR through MAGraph development, database modelling and reproducible Python/R/Snakemake workflows; thesis and catalogue-database contributions are detailed in the academic projects below.

CURRENT RESEARCH PROJECTS - ARONTIER

Industry research is described at a high level because it involves confidential work; internal datasets, implementation details and detailed unpublished findings are omitted.

Foundation-model development for multiplex tissue imaging

- Contribute to developing a foundation model for multiplex tissue images, aiming to connect molecular information with cellular and tissue context.
- Use a Vision Transformer and incorporate information from a protein language model; contribute to data preparation, model adaptation, pretraining and evaluation.

Computational analysis of multiplex tissue images

- Develop computational workflows connecting image processing, cellular measurements and phenotype review.
- Work with scientific collaborators to interpret tissue-derived measurements and retain reproducible, reviewable analyses.

Computational analysis of liver histology

- Develop and compare image-analysis methods for liver histology and steatosis.
- Examine reference-label quality and review model outputs with scientific collaborators, considering the level at which biological conclusions can be drawn.

ACADEMIC AND UNIVERSITY RESEARCH PROJECTS

MAGraph - master's thesis, TUM / AG Schirmer

- Designed and implemented a graph representation preserving assembly-derived gene-neighbourhood relationships, and evaluated a feedforward model for operon prediction.
- Integrated sequence, annotation and expression information; contributed thesis-derived work to the MetaGEAR preprint.

Joint scRNA-seq/scATAC-seq regulatory analysis - team project, LMU

- Contributed to a team project investigating cell-type-specific regulatory evidence from single-cell expression and chromatin-accessibility data in atherosclerosis.
- Selected expressed genes and transcription factors and linked reference-genome TF-binding motifs to nearby genes to construct candidate regulatory relationships.

Multi-cohort metagenomic catalogue database - university research internship, AG Schirmer

- Designed the initial MongoDB data model and query structure linking gene/protein sequences, functional annotations, abundance and cohort/sample metadata.
- Developed Python/Snakemake processing and BLAST/DIAMOND search workflows for cross-cohort biological queries.

Gene-family cluster refinement - supervised team project, AG Schirmer

- Co-developed and evaluated cluster-refinement methods using protein-family relationships and conserved genomic-neighbourhood context.
- Implemented Python workflow components and used reproducible metagenomic preprocessing and evaluation pipelines.

EnzymeMatch (EnzymeFinder) - bachelor's thesis, TUM

- Expanded and evaluated structure-based searches for substrate-specific biocatalysts using protein-ligand interaction patterns.
- Implemented geometric matching, candidate scoring and computational optimization in Python.

Protein intrinsic-disorder prediction - team project

- Contributed to a PyTorch-based workflow for residue-level regression from supplied protein-sequence embeddings.
- Worked on sequence preprocessing and model evaluation, comparing embedding sources and context-window choices.

PUBLICATIONS AND PREPRINT

Peer-reviewed publications

1. Hwang Y, **Cho WY**, Lee W, Joo I, Shin JI, Seo MR, Shin SH, Ko KS, Park KT, Chung YJ, Jung SH. Genomic signatures associated with epidemiologically defined high-risk pathogenic *Escherichia coli* isolates identified by interpretable machine learning. *Journal of Microbiology*. 2026;64(8):e2604011. doi:10.71150/jm.2604011.

2. Hong E, Shin Y, Kim H, **Cho WY**, Song WH, Jung SH, Lee M. PneusPage: a web-based tool for the analysis of whole-genome sequencing data of *Streptococcus pneumonia*. *Journal of Microbiology*. 2025;63(1):e2409020. doi:10.71150/jm.2409020.

3. Lee J, Shin JI, **Cho WY**, Park KT, Chung YJ, Jung SH. Genomic characteristics of *Vibrio vulnificus* strains isolated from clinical and environmental sources. *Genomics & Informatics*. 2024;22:26. doi:10.1186/s44342-024-00029-w.

4. Kim D, Shin JI, Yoo IY, Jo S, Chu J, **Cho WY**, Shin SH, Chung YJ, Park YJ, Jung SH. GenoMycAnalyzer: a web-based tool for species and drug resistance prediction for *Mycobacterium* genomes. *BMC Genomics*. 2024;25:387. doi:10.1186/s12864-024-10320-3.

Preprint

5. Jin S, Cenier A, Eisenhard L, Ríos E, Wetzel D, **Cho WY**, et al. Gene-centric metagenomic analyses reveal microbiome functional insights into diseases. *bioRxiv*. 2025. doi:10.1101/2025.09.29.679262.

TECHNICAL COMPETENCIES

- **Machine learning:** PyTorch, scikit-learn, transformer adaptation and self-supervised training, protein-sequence embeddings, multiple-instance learning, segmentation, feedforward networks, Random Forest, cross-validation and donor-level model evaluation.
- **Biological analysis and imaging:** IMC/CODEX, TIFF/OME-TIFF, Cellpose, marker quantification, spatial analysis, LLM-assisted phenotyping, metagenomics, bulk RNA-seq, scRNA-seq/scATAC-seq regulatory evidence and pathway enrichment.
- **Programming and reproducibility:** Python, R, shell, Git, Snakemake, SLURM, Docker, Apptainer and Linux administration.
- **Data and infrastructure:** MongoDB, MySQL, BLAST/DIAMOND, GCP Cloud Storage, biological database design, automated reporting, server access/storage/backup and job-queue management.

LANGUAGES

Korean: Native. **English:** Fluent; primary schooling in New Zealand; academic and professional use. **German:** Fluent; Austrian Gymnasium graduate (Matura); university studies in Germany.